

Project Documentation

Operational Resilience Analytics — Technical Reference

1. PROJECT OVERVIEW

1.1 Executive Summary

Analyzing a dataset of 1,500 incidents recorded between 2024 and 2025, this project evaluates operational resilience, recovery workflow efficiency, and vendor SLA compliance, applying a Star Schema data model and a rule-based data quality methodology.

1.2 Problem Statement

Context:

Operational incidents across multiple services and third-party vendors can generate significant financial loss and erode customer trust. This project analyzes incident-level data to surface root-cause patterns, quantify tolerance-limit exposure, and assess vendor reliability, in order to support more informed operational risk decisions.

Pain Points:

- Unclear Incident Classification: Confusion between technical issues, process delays, and logging errors due to a lack of standardized classification.
- Data Distortion from Outliers: Severe outliers skew aggregate metrics (e.g., Mean/Average).
- Monitoring Blind Spots: Difficulty detecting silent failures and data anomalies (impact without recorded downtime).

1.3 Business Objectives

- Reduce total downtime and financial loss.
- Assess vendor SLA compliance efficiency.
- Improve incident review and recovery workflows.

2. DATA SOURCE & ARCHITECTURE

2.1 Data Origin

- A synthetic dataset simulating data combined from ITSM Logs, Ticket System, and Gateway Logs.
- The dataset contains 1 fact table (Fact_Incidents) and 4 dimension tables (Dim_Vendor, Dim_Operation, Dim_Date, Dim_Tolerance_Limit), structured as a Star Schema.
- Source of data: generated via a Python script.
- Data size: 1,543 raw records, including 1,500 unique incidents, timeframe 1/1/2024 – 12/31/2025.

2.2 Data Schema & Key Attributes

Table Name	Type	Key Fields	Business Description & Purpose
Fact_Incidents	Fact	Incident_ID (PK), Limit_Key (FK), Vendor_Key (FK),	Stores core metric logs of operational disruptions, including Downtime_Minutes, Impacted_Txns_Count, Financial_Loss_AUD, and MTTR_Minutes (Mean Time to Repair).

		Date_Key (FK), Operation_Key (FK)	
Dim_Operation	Dimension	Operation_Key (PK)	Categorizes Critical Operations (Tier 1 vs. Tier 2) and assigns ownership to specific Business Units.
Dim_Tolerance_Limit	Dimension	Limit_Key (PK)	Holds the maximum acceptable Downtime and Failed-Transaction thresholds used to flag operational tolerance breaches (Max_Allowed_Downtime_Mins, Max_Allowed_Failed_Txns).
Dim_Vendor	Dimension	Vendor_Key (PK)	Captures Third-Party Service Provider details (SLA_Tier, Contract_Status) to support Vendor Risk oversight.
Dim_Date	Dimension	Date_Key (PK)	Enables time-series analysis across Month, Quarter, and Peak-Hour windows.

2.3 Operational Tolerance Limits

Each Critical Operation is assigned a Downtime and Failed-Transaction tolerance threshold, differentiated by Criticality Tier (Tier 1 services use stricter limits than Tier 2 services). These thresholds are used to flag incidents and operations that exceed acceptable service-disruption levels.

The modeling approach is inspired by general operational-resilience risk-appetite principles (in line with frameworks such as CPS 230), but the specific threshold values used here are illustrative and self-defined for the purpose of this project, not derived from real regulatory guidance.

Two breach concepts are used in this project:

- Single-Incident Breach: an individual incident where Actual Downtime exceeds the Max Allowed Downtime, or Impacted Transactions exceed the Max Allowed Failed Transactions.
- Cumulative Breach: the total Downtime for an Operation, summed across the reporting period, exceeds its Max Allowed Downtime — reflecting that tolerance limits are a period-based disruption budget, not a per-incident ceiling.

2.4 Data Quality & Limitations (Raw Data)

- Duplicated Incident_ID: 84 rows, corresponding to 42 unique Incident_IDs each appearing twice due to a simulated double-logging event.
- Missing Finance Info: 19 records.
- Negative Downtime: 29 records.
- Unclassified Vendors: 32 records.

3. METHODOLOGY & OPERATIONAL PATTERNS

3.1 Incident Classification Framework & Outlier Detection Logic

Pattern	Logic (Downtime vs MTTR)	Included in Downtime/Breach Stats?
Data Error - Invalid Downtime	dt < 0	Excluded - negative value is invalid
Outlier Downtime - Needs Data Audit	dt > IQR threshold	Excluded - suspected unit/entry error
Data Anomaly - Impact Without Downtime	dt = 0, MTTR = 0, txns > 0	Yes
Self-Resolved (Auto Recovery)	dt > 0, MTTR = 0	Yes

Successful Resolution	dt > 0, MTTR ≥ dt	Yes
Process Delay - Extended MTTR	admin_delay > IQR threshold	Yes (Downtime value still valid)
Logging Error - Premature Closure	dt > 0, MTTR < dt	Yes (Downtime valid; MTTR excluded from Admin Delay analysis only)
False Alert - No Actual Outage	dt = 0, MTTR > 0	Yes (no impact on Downtime SUM)

Outlier thresholds are calculated automatically using the IQR method (Interquartile Range, 1.5× multiplier) rather than a fixed cutoff, so the threshold adapts to the actual distribution of the data.

3.2 Data Exclusion & Inclusion Rules

Is Valid For Stats	Operational Pattern	# Incidents	% of Incidents
True	Successful Resolution	931	62.07%
True	Self-Resolved (Auto Recovery)	182	12.13%
False	Logging Error - Premature Closure	96	6.40%
True	Process Delay - Extended MTTR	95	6.33%
False	Outlier Downtime - Needs Data Audit	77	5.13%
True	Data Anomaly - Impact Without Downtime	47	3.13%
True	False Alert - No Actual Outage	43	2.87%
False	Data Error - Invalid Downtime	29	1.93%
	Total	1,500	100.00%

Note: Logging Error - Premature Closure retains a valid Downtime value and is therefore included in Downtime/Breach statistics; it is excluded only from Admin Delay analysis, where MTTR reliability specifically matters.

3.3 Vendor Risk Score Model

Formula: Min-Max Normalized Weighted Score.

Rationale: Min-Max normalization was chosen over a simple Max-based ratio because it is less sensitive to a single extreme vendor value distorting the scale for all other vendors, while remaining simpler to compute and explain than a Z-score approach.

Score component = (Current Value – Min) / (Max – Min), where Min and Max are taken across all vendors for that metric.

Weighted components:

- Financial Loss Score — weight 40%
- Incident Count Score — weight 20%
- Average Downtime Score — weight 25%
- P95 Downtime Score — weight 15%

Example — Financial Loss Score:

- Min Historical Loss = \$0
- Max Historical Loss = \$100,000
- Vendor A Loss = \$25,000
- Financial_Loss_Score = (\$25,000 – \$0) / (\$100,000 – \$0) = 0.25
- Weighted contribution = 0.25 × 40% = 10 points

Final Output: Risk Score ranges from 0 (Low Risk) to 100 (High Risk), summing the weighted contribution of all four components.

4. DASHBOARD ARCHITECTURE & USER GUIDE

- Page 1: Operational Pattern Definitions & Data Exclusion Rules.
- Page 2: Executive Overview & Baseline Metrics.
- Page 3: Core Performance vs. Admin Delay Breakdown.
- Page 4: SLA Breach & Operational Tolerance Analysis.
- Page 5: Vendor Risk Profile & Performance Ranking.
- Page 6: Outlier Deep-Dive (Data Anomaly & Logging Errors).

5. DATA QUALITY GOVERNANCE & AUTOMATED RULES

To ensure long-term data integrity and prevent metric distortion, the following automated Data Quality (DQ) rules are designed to be integrated into the ETL process (Power Query / pipeline):

1. Negative Value Scrubbing (Invalid Downtime)

- Rule: Reject or isolate any record where `Downtime_Minutes < 0`.
- Action: Flag as Data Error and route to the IT Logging Team for system clock sync audit.

2. Outlier Detection & Isolation

- Rule: Flag records where `Downtime_Minutes > Q3 + 1.5 × IQR` within each Operation category.
- Action: Exclude automatically from core SLA/Executive dashboards; route to the Outlier Deep-Dive view for manual audit.

3. Data Anomaly & Silent Failure Tagging

- Rule: Identify records where `Downtime_Minutes = 0`, `MTTR_Minutes = 0`, but `Impacted_Txns_Count > 0`.
- Action: Categorize as Data Anomaly to track monitoring gaps where outages occur without formal logging.

4. Premature Closure Validation

- Rule: Detect records where `MTTR_Minutes < Downtime_Minutes (dt > 0)`.
- Action: Keep `Downtime_Minutes` for outage calculations, but flag `MTTR` as untrusted (automatically excluded from Admin Delay metrics).

6. LIMITATIONS

- No granular timestamp: the dataset only records `Is_Peak_Hour` (binary) and `Date_Key` (day-level); there is no actual time-of-day field. Time-based analysis (e.g., business hours vs. off-hours) is therefore approximated, not precise.
- Financial Loss is derived from Downtime, not independently measured: in this synthetic dataset, `Financial_Loss_AUD` and `Impacted_Txns_Count` are both generated as a function of `Downtime_Minutes`. This means outlier Downtime records (suspected data-entry errors) proportionally inflate Financial Loss as well — the two metrics are not statistically independent in this dataset, which would not be the case with real production data.
- No designed seasonality: incident dates were randomly distributed across the two-year period; the dataset does not simulate real-world patterns such as payday spikes or end-of-quarter surges.

- Small sample size at the vendor/tier level: some SLA tiers (e.g., Bronze, Platinum) are represented by only one vendor, which limits the statistical confidence of any tier-level conclusion.
- Synthetic data only: all figures are illustrative. This project is intended to demonstrate analytical methodology and dashboard design, not to represent validated real-world findings.